



Cement-based materials: Pozzolanic activities of mineral additions are compromised by the presence of reactive oxides

R. Berenguer^{a,b,*}, N. Lima^{a,c}, L. Pinto^b, E. Monteiro^{c,d}, Y. Povoas^c, R. Oliveira^d, N.B. D. Lima^{a,b,**}

^a National Institute of Union Technology and Coating of Materials, Federal University of Pernambuco, 50740-540, Recife, PE, Brazil

^b Department of Fundamental Chemistry, Federal University of Pernambuco, 50740-540, Recife, PE, Brazil

^c Polytechnic School, University of Pernambuco, 50720-001, Recife, PE, Brazil

^d Department of Civil Engineering, Catholic University of Pernambuco, Recife, PE, 50050-900, Brazil

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ABSTRACT

Presence of reactive oxides, such as sodium, potassium, and magnesium oxides, compromise the pozzolanic activities of mineral additions to cement-based materials. That is because these oxides react with water to form hydroxides. Subsequently, these hydroxides may react with the otherwise non-reactive oxides, such as silicon and aluminum oxides, to form various silicate and aluminate phases in the cement-based material, that finally compete with the desired CSH phase. Further, the strong pozzolanic activity of the silica fume is seemingly due to its large ability to form the calcium silicate hydrated systems when compared with both metakaolin and fly ashes. All these results have been established by X-ray fluorescence, microstructural, and chemometric analyses that were applied to investigate the pozzolanic activities, thermodynamic, structural, and mechanical properties of a set of mineral systems widely applied to cement-based materials (silica fume, metakaolin, and fly ashes). The chemometric analyses revealed that the materials prepared with silica fume exhibited the largest mechanical properties, whereas the reference the lowest. The cement-based materials with metakaolin and fly ashes present intermediary resistances and similar behaviors. In conclusion, strict control of the purity of the material always needs to be performed to obtain mineral systems with powerful pozzolanic activities.

1. Introduction

Pozzolanic activities are target properties of mineral systems that can be added to cement-based materials [1–3]. A pozzolanic material is defined as a system that reacts with calcium hydroxide forming compounds that exhibit cementitious characteristics [4,5]. These pozzolanic materials are widely employed in cement-based materials to increase their performance, such as the mechanical properties [6–9] and the resistance to aggressive external agents [10–14].

Several methods were reported in order to quantify the pozzolanic activity of the cement-based materials [15–17]. In particular, the modified Chapelle's method is considered a simple strategy to quantify this activity. The modified Chapelle's method [17] evaluates the kinetic aspects of the pozzolanic reactions between calcium hydroxide and the investigated mineral system. Edmeades et al. [18] reported that the kinetic properties of the pozzolanic reactions depend on the specific

surface area of the investigated material, as well as on the ratio between the quantity of the water and solid material.

Recently, Soares et al. [4] reported aspects of the kinetic and thermodynamic properties associated with pozzolanic activities of three types of nanoparticles of silica by using a method based on electronic sensors of pH. The authors concluded that their optimizations on Chapelle's method were useful to reveal aspects of the complex physical-chemistry of the reactions involving components of the cementitious materials [4]. A strategy to increase the pozzolanic activity of the granulated copper slag was also reported by Feng et al. [3]. The authors reported that the presence of calcium oxide in the mixture of granulated copper slag is capable of increasing its pozzolanic activity [3]. Although important advances were reported revealing aspects of the chemistry of the pozzolanic activities of mineral additions [19,20], more aspects also need to be clarified, for example: the physico-chemical properties of the chemical reactions associated with the pozzolanic

* Corresponding author. National Institute of Union Technology and Coating of Materials, Federal University of Pernambuco, 50740, Recife, PE, Brazil.

** Corresponding author. National Institute of Union Technology and Coating of Materials, Federal University of Pernambuco, 50740, Recife, PE, Brazil.

E-mail addresses: romildo.berenguer@ufpe.br (R. Berenguer), nathalia.bluma@ufpe.br (N.B.D. Lima).

activities by considering a molecular perspective.

In the present work, new insights into the complex chemical nature of the pozzolanic activities of a set of mineral systems are advanced. The investigated mineral systems are silica fume, metakaolin, and fly ashes. Further, the XRF spectrometry and thermogravimetric analyses are employed to evaluate thermodynamic aspects of the hydration reactions associated with the processing of the cement-based materials prepared with mineral additions. Furthermore, structural, and microstructural aspects of the materials are studied by x-ray diffraction (XRD), scanning electron microscope (SEM), and energy-dispersive x-ray spectroscopy (EDS) experiments. Finally, insights into mechanical performances (axial compressive strength and tensile resistances) of the cement-based materials are investigated by a chemometric approach.

2. Experimental

2.1. Materials

The following materials were employed: Portland cement of type III of ASTM [21]; (source tecnosil); silica fume (source tecnosil); metakaolin (source HP Ultra); fly ashes (local source); sand standardized by the Institute of Technological Research of Brazil in accordance with NBR 7214 (2015) [22]; gravel #19 (local source); and distilled and deionized water.

2.2. Modified Chapelle's method

According to NBR 15895 (2010) [17], to perform the modified Chapelle's method, 1g of the mineral system (silica fume, metakaolin, or fly ashes) and 2g of calcium oxide were added to 250 mL of distilled/deionized water, in an Erlenmeyer heated to $90 \pm 5^\circ\text{C}$, that was left stirring for 16h. Simultaneously, the same procedure was performed to determine the standard test containing only a solution of 1g of calcium oxide in 250 mL of distilled/deionized water. Subsequently, a solution of sucrose in 250 mL of water was prepared that was added to the Erlenmeyer with the water solution of the investigated material evaluated. This mixture was stirred for 15 min. Further, 50 mL of this mixture was titrated with 0.1 M hydrochloric acid solution by using a 1 g/L of a solution of phenolphthalein solution as an indicator. The same procedure was performed for the standard solution. In order to quantify the pozzolanic activity index (I_{PA}) by the modified Chapelle's method, the following expression can be employed: $I_{PA} = \frac{28 \times (V_1 - V_2) \times F_c}{m_1}$, where V_1 is the volume of the 0.1 M HCl solution employed in the test of the standard solution, V_2 is the volume of the 0.1 M HCl solution employed in the test of the solution with the investigated system, m_1 is the mass of the investigated material, and F_c is 1.32 (correction factor for the usage of the 0.1 M HCl solution [17]).

2.3. Processing of the cement-based materials

The samples of the reference material were prepared by employing the proportion between Portland cement, sand, and water of 1:3:0.48. In this sense, 0.624 kg of Portland, 1.872 kg of sand, and 0.300 kg of water were mixed in a plaster/mortar electromechanical mixer I 3010 Contenco. For all materials, the water/cement ratio was 0.48. Subsequently, the samples of the cement-based materials with mineral additions (silica fume, metakaolin, and fly ashes) were prepared by employing the same quantities of the cement, sand, and water of the reference, and, with 93.6 g of the mineral system. This quantity of mineral addition corresponds to 15% of the cement employed. Further cylindrical samples of the cement-based materials were prepared, where their dimensions were 5 cm $\times$ 10 cm. Finally, the following curing times were considered: 28, 63, 91 and 365 days. All cement-based materials prepared were left to submerged curing. After the curing times, the cement-based materials were dried and their structural, morphological, thermal, and mechanical

properties were evaluated.

2.4. Characterizations

2.4.1. XRF spectrometry

The samples were prepared in a Shimadzu equipment, where a load of 150 kN was applied over a period of 1 min. Subsequently, the samples were dried at a temperature of 90°C . Further, the proportions between the inorganic compounds present in the cement-based materials were identified using a Shimadzu equipment model XRF 1800.

2.4.2. Thermogravimetric analysis

The thermogravimetric analyzes were performed in an equipment STA 449 F3 JUPITER NETZSCH. The conditions employed were inert atmosphere, maximum flow of 100 ml/min and heating range from 30°C to 1000°C , with a heating rate of $10^\circ\text{C}/\text{min}$, with sample mass of $20 \text{ mg} \pm 0$ and purging at $50^\circ\text{C}/\text{min}$.

2.4.3. Experiments of water absorption capacity and SEM/EDS

The cylindrical samples were immersed in water for the following times: 3h, 6h, 24h, 48h, and 72h. All samples had their masses measured after each time considered, and the values of water absorption capacity ($A_{\%}$) were calculated by using the equation $A_{\%} = \frac{M_{sat} - M_d}{M_d} \times 100\%$, where M_{sat} is the mass of the saturated sample after immersion in water and M_d is the mass of the dried sample.

Further, the SEM analyses were carried out in a scanning electron microscope TESCAN, model MIRA3, with 10 kV voltage and SE detector. The samples were placed in a Stub with a carbon ribbon for image acquisition by using metallization and magnitude (MAG) ranging from $650\times$ to $20k\times$. Furthermore, the SEM/EDS analyses of the cement-based materials were performed by using a detector of x-ray energy-dispersive spectroscopy EDS OXFORD (X-Maxⁿ).

2.4.4. XRD experiments

For the x-ray experiments of the samples, a D8 Advance diffractometer Bruker was used. A copper tube with a wavelength (λ) of $1.54060 \text{ \AA}$, current of 10 mA and voltage of 30 kV was also employed. The XRD experiments were performed with 2θ range from 7° to 80° , time of 1s, and increment of 1° . In addition, the speed angular step of $0.02^\circ/\text{s}$ and a divergent gap of 1° were considered. The phases present in the samples were identified by using the software DIFFRAC plus-EVA, with a database centered on the Crystallography Open Database system. Further, the samples of cement-based materials were studied by using the Rietveld refinement method, where the TOPAS software and Powder Diffraction Files were employed.

2.4.5. Mechanical properties

The mechanical properties of the axial compressive strength of the cement-based materials were measured with samples with their surface dried. The axial compressive strength measurements were carried out with the load application speed of 0.4 MPa/s . The corresponding value employed in the experiments of tensile splitting strength was 0.05 MPa/s . Both mechanical properties of the cement-based materials were measured in a 200t hydraulic press SOLOTEST. For each type of mechanical property, six samples were considered.

2.5. Chemometric analysis

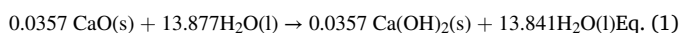
A large quantity of data of mechanical properties was acquired by varying the type of material and time parameters, while the responses were measured by axial compressive strength and tensile resistance experiments. These were performed in a manner that so that it was possible to evaluate all 96 experiments systematically, for each response, in interpretable tables and graphics using chemometrics modeling. The data were compiled in order to identify the influence of

each of the parameters and their interactions using the design of experiments (DOE) tool to carry out the analyses with a multilevel categorical modeling [23]. For each class of measurements, a response surface that graphically describes the least-squares model was built and employed to identify the parameters that give the higher axial compressive strength and the tensile resistance individually [23]. The response function was built by using the Derringer and Suich criteria by identifying the parameters that result in the highest value for both responses simultaneously [23].

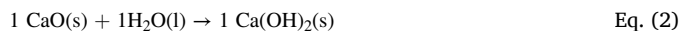
3. Results and discussion

3.1. The pozzolanic activities of the mineral systems

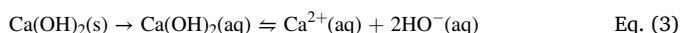
By employing the modified Chapelle's method, the values of the pozzolanic activity indices (I_{PA}) of the mineral systems studied were 1498 mgCaO/g of silica fume, 1195 mgCaO/g of metakaolin, and 380 mgCaO/g of fly ashes. As expected, both silica fume and metakaolin exhibited a strong pozzolanic activity when compared with the corresponding value of the fly ashes, that revealed a poor pozzolanic activity. The modified Chapelle's method is composed of comparisons between a reference sample (an aqueous solution of the calcium oxide) and an investigated solution (an aqueous solution of a mixture of calcium oxide and the material suspected to have pozzolanic activity). From a molecular perspective, the hydration of calcium oxide in the reference sample is (2g of CaO in 250 mL of water):



Equation (1) can be interpreted as the 1:1 hydration reaction between calcium oxide and water:



By considering the cement nomenclature, the chemical reaction of Eq. (2) is $C + H \rightarrow CH$. The hydration reaction of the calcium oxide is a spontaneous process in which the Gibbs energy ($\Delta_r G = -57.0 \text{ kJ/mol}$) is governed by the enthalpy of reaction ($\Delta_r H = -64.5 \text{ kJ/mol}$) [24]. When the calcium hydroxide is an aqueous medium, a chemical equilibrium can occur:

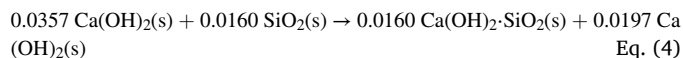


The chemical equilibrium showed in Eq. (3) can be associated with the presence of all species in the aqueous medium formed in the reference sample of the modified method of Chapelle. The partial dissolution of calcium hydroxide in an aqueous medium is due to its poor solubility, and an equilibrium between the ionic compound Ca(OH)_2 , and its corresponding cationic (Ca^{2+}) and anionic (HO^-) species tends to be formed. The first step of the dissolution of calcium hydroxide in water is a non-spontaneous process ($\Delta_r G = +187.4 \text{ kJ/mol}$), and not energetically favorable ($\Delta_r H = +212.1 \text{ kJ/mol}$). This non-spontaneous reaction occurs because the calcium hydroxide is an ionic solid, and its dissolution leads to cleavages of the ionic bonds, typically endothermic processes. The species Ca(OH)_2 , Ca^{2+} , and 2HO^- in the aqueous medium are capable of interacting with the water molecules of the solvent. The calcium hydroxide is capable of forming stable hydrogen bonds [24], and the ions Ca^{2+} and HO^- are solvated by the water solvent [25]. The standard sample employed in the modified Chapelle's method can be considered as a mixture that contains, essentially, calcium hydroxide and its corresponding ions in an aqueous medium.

A molecular perspective of the behavior of both calcium oxide and calcium hydroxide in the aqueous medium may be useful to establish physico-chemical aspects of the role of the reference sample of the modified Chapelle's method. The use of the modified Chapelle's method to detect pozzolanic activities of mineral additions is essentially to evaluate a mixture of 2g (0.0356 mol) of the calcium oxide and 1g of the investigated system that is added to an aqueous medium. All solids

investigated in this article are essentially composed of inorganic materials, such as oxide and hydroxide compounds, and molecular aspects of the pozzolanic activities of mineral systems are now elucidated. Table 1 shows the XRF data for all oxide compounds that compose the mineral systems and the Portland cement.

From Table 1, the main component of the silica fume is silicon dioxide (~96%), and when 1g of the mineral system is added to the reference solution, the following reaction can occur:

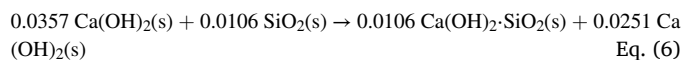


That is equivalent to:

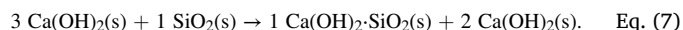


An important aspect of the silica fume that exhibits an exacerbated pozzolanic activity is its more significant purity in terms of silicon oxide (~96%), leading to a lower impact on other oxide compounds that may compete with the transformation of this material into calcium silicate hydrate phases; phases that naturally exhibit properties of cementitious materials. Therefore, the strong index of pozzolanic activity of the silica fume (I_{PA} of 1498 mgCaO/g of silica fume) measured by the modified Chapelle's method is seemingly due to its large ability to form the calcium silicate system when compared with the mineral additions that exhibit lower pozzolanic activities. In this sense, the metakaolin pozzolanic activity is now evaluated.

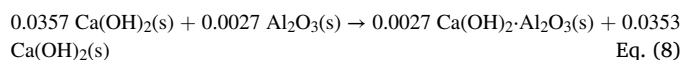
The metakaolin is essentially composed of SiO_2 (~64%) and Al_2O_3 (~28%), which are 92% of this mineral system (Table 1). Both silicon dioxide and aluminum oxide can react with calcium hydroxide present in the standard solution employed in the modified Chapelle's method. The silicon dioxide present in 1g of metakaolin (63.92%, i.e., 0.0106 mol in 1g of the material), leads to the following balanced main global reaction:



which is equivalent, in integer numbers, to:



Equation (7) reveals that silicon dioxide is the limiting reactant, that can lead to a more capability of this inorganic compound to react with calcium hydroxide. The quantity of aluminum oxide (28.01%, i.e., 0.0027 mol in 1g of the metakaolin) is also capable of leading to an alternative reaction pathway:



equivalent to:



Studies revealed that the formation of calcium aluminate hydrated phases is estimated to be an spontaneous process [26]. More complex

Table 1
XRF spectrometry data of the silica fume, metakaolin, fly ashes, and Portland cement.

Inorganic Compound	Silica Fume	Metakaolin	Fly ashes	Portland Cement
SiO_2	95.56%	63.92%	70.11%	18.92%
CaO	0.29%	0.14%	1.58%	60.59%
MgO	0.52%	1.06%	0.81%	4.74%
Fe_2O_3	0.81%	2.55%	5.24%	2.69%
SO_3	0.40%	0.05%	0.50%	2.86%
K_2O	0.96%	2.63%	2.39%	0.32%
Al_2O_3	0.12%	28.01%	17.34%	5.35%
Na_2O	0.95%	0.07%	0.66%	0.28%

phases can be formed when the metakaolin system is added to the standard solution of modified Chapelle's method [27]. For example, a recent work revealed that the CASH phase is formed by spontaneous processes [26].

An alternative reaction pathway involving the metakaolin could be from the hydration of Al_2O_3 rather than the hydration of calcium oxide in the standard sample of the modified Chapelle's method. However, the hydration of aluminum oxide, contrary to the hydration of calcium oxide, is not energetically favorable to occur because the enthalpy of the hydration reaction is $+3.0$ kJ/mol. The corresponding Gibbs energy is only -2.0 kJ/mol. These thermodynamic quantities indicate that the hydration of aluminum oxide leads to an equilibrium situation, as previously reported [24].

The main difference between the pozzolanic activity of the silica fume and the metakaolin systems, is apparently due to the possibility of the formation of more diverse phases when the metakaolin is considered instead of silica fume.

On the other hand, by evaluating the chemical compositions of the metakaolin with the fly ashes, both are, essentially, composed of SiO_2 and Al_2O_3 (Table 1). However, their corresponding pozzolanic activity values are different: 1195 mgCaO/g of metakaolin and 380 mgCaO/g of fly ashes. In this sense, more chemical aspects also need to be considered when the mineral additions are originally prepared by heating systems that are discarded by industry, such as the fly ashes. The same may be extended to other types of ashes, such as ashes of sugarcane bagasse [28, 29] and rice husk [30–32]. As can be seen from Table 1, the fly ashes present more amounts of oxide compounds that can perform competitive hydration reactions when compared to both silica fume and metakaolin. These oxide compounds are CaO, MgO, K_2O , and Na_2O . From Table 1, for the silica fume and metakaolin employed, these oxide compounds are, respectively, 2.72% and 3.90% of their chemical compositions. The corresponding value for the fly ashes is 5.49%. By comparing both metakaolin and fly ash systems, there exists an increase of 41% of oxide compounds that are able to compete with the hydration of the free calcium oxide present in the investigated solution of the modified Chapelle's method. The alternative hydration reactions are shown in Eqs. (10)–(12).



The $\Delta_r G$ values of the hydration reactions showed in Eqs. (10)–(12) are -27.0 kJ/mol; -130.4 kJ/mol; and -146.7 kJ/mol, respectively. The corresponding and $\Delta_r H$ values are -37.1 kJ/mol; -200.2 kJ/mol; and -150.9 kJ/mol. All these hydroxide compounds in an aqueous medium may enable competitive reactions that can lead to a reduction of the formation of the calcium silicate hydrated system. If there are more possibilities of hydration reactions when the Chapelle's method is performed, several alternative hydrated phases can be formed, such as magnesium silicates, potassium silicates, potassium aluminates, sodium silicates, and sodium silicate aluminates. More possibilities of formation of different types of phases may reduce the pozzolanic activities of a mineral system, such as the fly ashes. Therefore, the nature of the pozzolanic activities of mineral systems is associated with the possibilities of formation of silicate and aluminate phases that are enabled by the hydration of a set of oxide systems. If a strong pozzolanic material is targeted, a higher selectivity of the phases that can be formed must be considered. Therefore, strict control of the purity of the material always needs to be performed to obtain powerful pozzolanic activities.

3.2. Thermodynamic aspects of the hydration reactions of the cement-based materials

Because the hydration reactions associated with the processing of the

cement-based materials involve simple inorganic systems that have their thermodynamic properties very well established [33], the values of the Gibbs energy ($\Delta_r G$) and enthalpy ($\Delta_r H$) of these reactions can be quantified, as shown in Table 2. From Table 2, all hydration reactions associated with the processing of the cement-based materials are similar. These reactions are very spontaneous processes ($\Delta_r G$ average -398.2 kJ/mol), where the corresponding enthalpy of reactions, $\Delta_r H$, average -455.2 kJ/mol. Therefore, the thermodynamic properties revealed that the mineral additions considered are not capable of affecting, in an important manner, the hydration of the Portland cement when the materials were prepared. All details of this thermodynamic analysis are present in the Supplementary Material.

Further, to evaluate other energetic aspects of the cement-based materials under high temperatures, thermogravimetric analysis was performed (Fig. 1). The thermal analysis revealed that all cement-based materials prepared with mineral additions exhibited loss of mass at temperatures between 400°C and 450°C . This behavior was reported in a previous article [34], and is associated with the portlandite phase. The values of the loss of mass of the cement-based materials in temperatures between 400°C and 450°C are ordered as follows: (1) cement-based material of reference; cement-based material concrete with (2) fly ashes; (3) metakaolin and (4) silica. This order reveals that the mineral additions can interact with the main components of the composite, which tend to stabilize the material in an energetic perspective. By considering both types of solid ingredients used in the cement-based materials (mineral additions and the Portland cement), no mass losses were observed in 400°C and 450°C . This reinforces the possibility of chemical interactions between the portlandite and the mineral systems. From Fig. 1, a poor decomposition of the materials by carbonation processes was also verified when the samples are at a temperature between 700°C and 800°C . The corresponding values of loss of mass ($>1\%$) indicated that the Portlandite was capable of reacting with CO_2 .

3.3. Structural properties

In order to evaluate structural aspects of the cement-based materials, Fig. 2 shows XRD patterns of both ingredients and the cement-based materials. From Fig. 2, signals of both crystal structures and amorphous phases of the materials are present. The Portland cement has several phases, in which the main signals appear at the following 2θ values: 12.1° (CaSO_4 phase); 29.4° and 32.1° (C_2S phase); 30.0° and 32.5° (C_3S phase); and 33.5° (C_3A phase) and 34.4° , and C_4AF (C_4AF phase). All mineral additions (silica fume, metakaolin, and fly ashes) exhibited amorphous phases. For these inorganic systems, a large

Table 2

Values of the Gibbs Energy ($\Delta_r G$) and enthalpy ($\Delta_r H$) of the main hydration reactions associated with the processing of the four types of cement-based materials studied (reference, silica fume, metakaolin, and fly ashes cement-based materials). The physico-chemical data were considered at 298 K.

Hydration Reactions	$\Delta_r G$ (kJ. mol^{-1})	$\Delta_r H$ (kJ. mol^{-1})
6.74 CaO(s) + 0.74 MgO(s) + 0.02 K ₂ O(s) + 0.03 Na ₂ O(s) + 16.65H ₂ O(l) → 6.74 Ca(OH) ₂ (s) + 0.74 Mg(OH) ₂ (s) + 0.04 KOH(s) + 0.06 NaOH(s) + 9.07H ₂ O(l)	−399.7	−457.0
6.75 CaO(s) + 0.75 MgO(s) + 0.03 K ₂ O(s) + 0.04 Na ₂ O(s) + 16.65H ₂ O(l) → 6.75 Ca(OH) ₂ (s) + 0.75 Mg(OH) ₂ (s) + 0.06 KOH(s) + 0.08 NaOH(s) + 9.01H ₂ O(l)	−399.1	−456.4
6.74 CaO(s) + 0.76 MgO(s) + 0.05 K ₂ O(s) + 0.03 Na ₂ O(s) + 16.65H ₂ O(l) → 6.74 Ca(OH) ₂ (s) + 0.76 Mg(OH) ₂ (s) + 0.10 KOH(s) + 0.06 NaOH(s) + 8.99H ₂ O(l)	−398.0	−456.1
6.77 CaO(s) + 0.76 MgO(s) + 0.05 K ₂ O(s) + 0.04 Na ₂ O(s) + 16.65H ₂ O(l) → 6.77 Ca(OH) ₂ (s) + 0.76 Mg(OH) ₂ (s) + 0.10 KOH(s) + 0.08 NaOH(s) + 8.94H ₂ O(l)	−394.2	−451.1

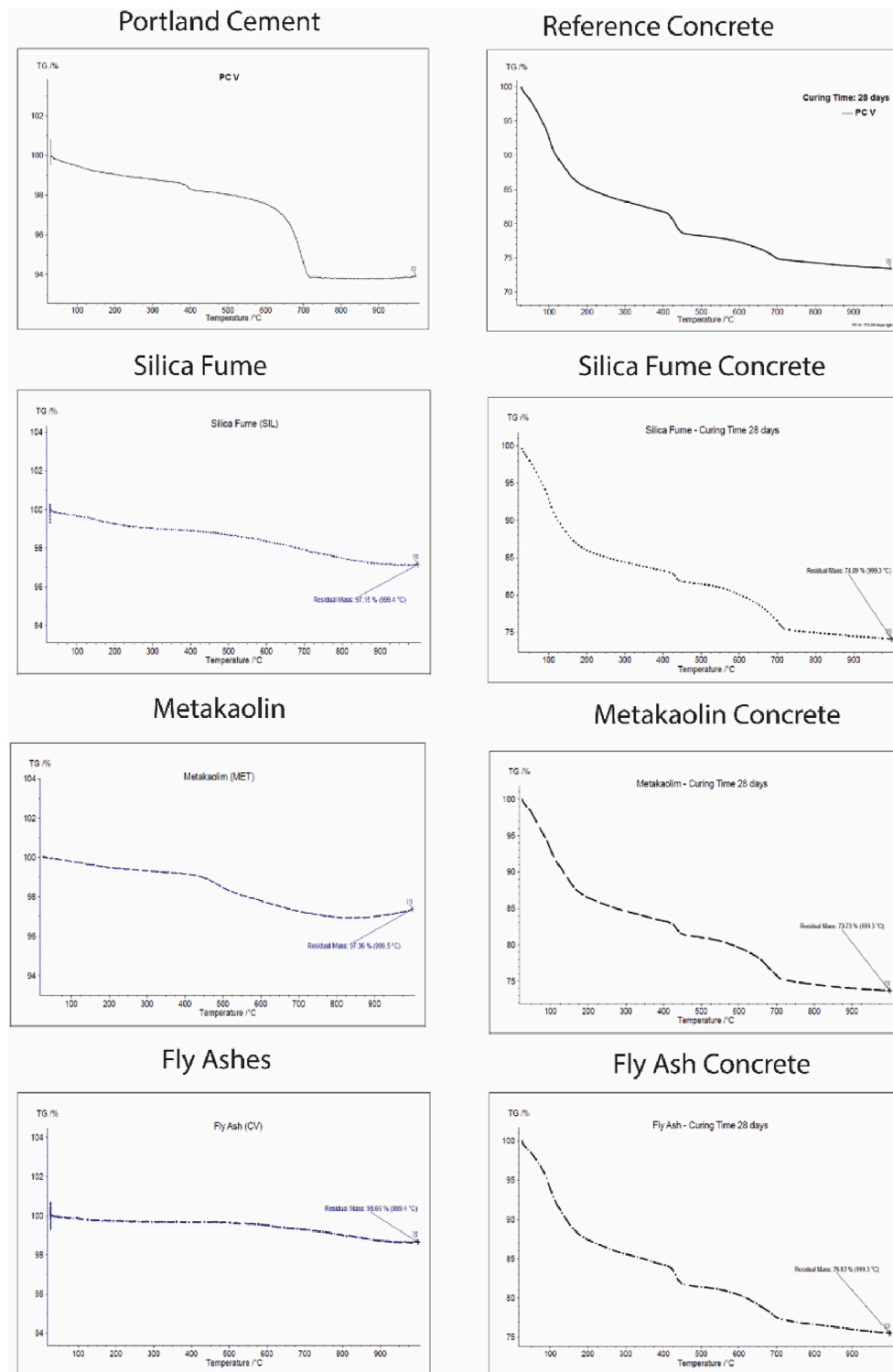


Fig. 1. Thermogravimetric data of the precursor solid materials and of the cement-based materials.

amount of the amorphous phase was detected for the silica fume.

On the other hand, when the cement-based materials were prepared, some phases lose intensity, and others are transformed into new phases. For example, a characteristic peak at 9.1° is associated with the formation of the ettringite phase ($\text{Ca}_6\text{Al}_2(\text{SO}_4)_3(\text{OH})_{12}$). Two characteristic peaks at 18° and 34° are verified, that correspond to the portlandite phase ($\text{CA}(\text{OH})_2$). These phases are formed by the hydration of precursor phases present in the Portland cement, such as the calcium silicates $3\text{CaO} \cdot \text{SiO}_2$ (C_3S) and $2\text{CaO} \cdot \text{SiO}_2$ (C_2S). The important amorphous

phase *CSH* was detected between 28° and 33° .

Further, when the systems of silica fume, metakaolin, and fly ashes are added to the cement-based materials, similar structural aspects were verified. For example, all samples present the characteristic phases when compared to the reference system, but with lower intensities. The portlandite phase was also detected in the signals between 18° and 34° . The XRD patterns of the cement-based materials with mineral additions also revealed that the formation of the *CSH* phase appears between 28° and 33° . The mineral addition contributes to the formation of the *CSH* phase

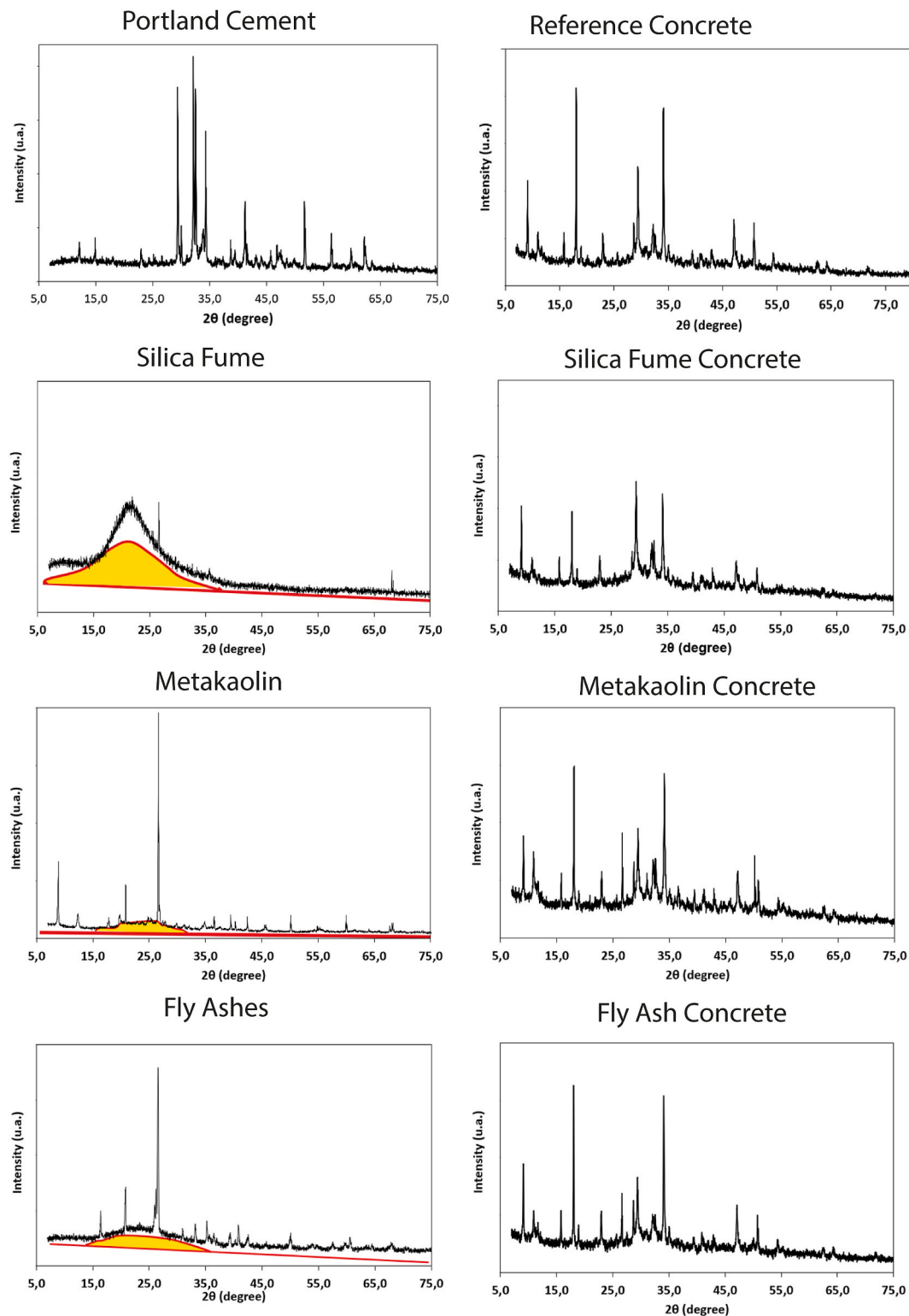


Fig. 2. X-ray diffraction patterns of the precursor and cement-based materials.

when the C_2S and C_3S phases are transformed by hydration processes. This is because the pozzolanic activities of the mineral systems result from the reaction of silicon dioxide with the portlandite phase forming the hydrated calcium silicate phase. The ability of the mineral systems to react with the portlandite phase was reported by Hoppe Filho et al. [35] and Berenguer et al. [34]. The amorphous phase of the samples affects the interactions between the components of these materials, in which there are changes in the crystal lattices present in the cement-based

material.

Furthermore, mineral phases were identified in all investigated cement-based materials (Fig. 3). The formation of the ettringite phase (9.1°) is observed for all samples. The crystalline quartz phase was identified at 26.5° for materials prepared with metakaolin and fly ashes. The signals associated with the calcite were also identified for all samples in 29° . For values of 2θ between 28° and 33° the phases of hydrated calcium silicate and calcium silicate were identified.

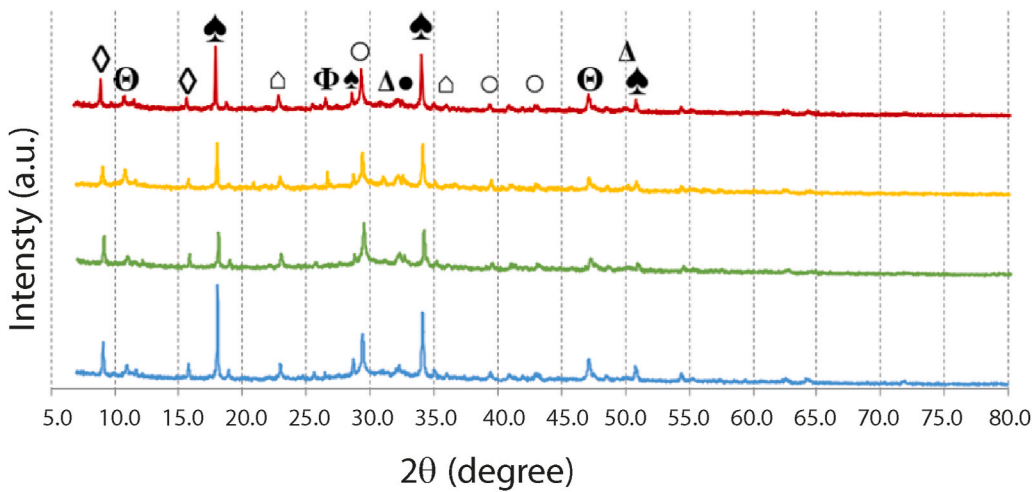


Fig. 3. X-ray diffraction patterns of the cement-based materials by considering the Rietveld refinement data. The highlighted codes in the diffractograms correspond to the following Powder Diffraction Files (PDFs): $\bullet$ PDF 44–1481 $\text{Ca}(\text{OH})_2$ (Portlandite), $\diamond$ PDF 41–1451 $\text{Ca}_6\text{Al}_2(\text{SO}_4)_3(\text{OH})_{12}\cdot 26\text{H}_2\text{O}$ (Ettringite), $\circ$ PDF 05–0586 CaCO_3 (Calcite), $\bullet$ PDF 49–1673 Ca_2SiO_4 (Calcium Silicate), Δ PDF 33–0306 $\text{Ca}_{1.5}\text{SiO}_{3.5}\cdot x\text{H}_2\text{O}/1.5\text{CaO}\cdot\text{SiO}_2\cdot x\text{H}_2\text{O}$ (Calcium Silicate Hydrate), $\circ$ PDF 41–0219 $\text{Ca}_4\text{Al}_2\text{O}_6\text{CO}_3\cdot 11\text{H}_2\text{O}/3\text{CaO}\cdot\text{Al}_2\text{O}_3\cdot\text{CaCO}_3\cdot 11\text{H}_2\text{O}$ (Aluminate), Θ PDF 14–0698 $\text{Al}_{13}\text{Si}_5\text{O}_{26}(\text{OH}, \text{F})_{18}\text{Cl}$ (Sorosilicate mineral), and Φ PDF 86–1560 SiO_2 (Quartz). The samples of the four types of cement-based materials are destined as, reference (blue) and with additions of silica fume (green), metakaolin (yellow) and fly ash (red). (For interpretation of the references to

colour in this figure legend, the reader is referred to the Web version of this article.)

By considering the x-ray diffraction patterns of the cement-based materials (reference and the materials prepared with additions of silica fume, metakaolin, and fly ashes), the main inorganic phases were investigated by using the Rietveld refinement, as can be seen in Table 3.

From Table 3, more structural aspects were identified, especially for the significant quantity of the amorphous systems formed after 91 days of curing time. By considering the ettringite phase, it was verified that all samples showed similar values, where the quantity of plaster present in Portland cement leads to the formation of this phase. Besides, mineral additions did not lead to the formation of other phases of ettringite (9.1°) in a late curing time.

For the materials prepared with metakaolin and fly ashes, the quartz crystalline phase (SiO_2) was identified at 26.5° . These results agree with the analyses previously carried out by using the XRF spectrometry, where these two samples present large amounts of silica material (SiO_2), 63.92% and 70.11%, respectively. This reveals that the silicon dioxide is present in its crystalline form. However, it was also identified that the quantity of the cement-based materials with silica fume (95.56%) did not display the quartz phase. This shows that the silica fume added is in an amorphous form.

All samples with mineral additions obtained increased the proportion of the amorphous phase when compared to the reference material. This indicates that mineral additions intensified the occurrence of this phase when compared with the reference data. It was also verified that all samples with mineral additions had a lower amount of portlandite phase (18° and 34°), and, seemingly, this occurred because the portlandite phase was consumed by the mineral additions generating the pozzolanic reactions, which are direct consequences when the proportions of amorphous phases are increased. From Table 3, the amorphous phases detected in the samples indicate, mainly, the extent of CSH formation. These phases in the samples of Portland cement, as well as in the samples of fly ashes, metakaolin, and silica fume, are, respectively, 27.91%, 35.92%, 37.55%, and 41.19% of the solids. The presence of CSH improves the mechanical properties of cement-based materials. In short, the addition of SCMs increases the amorphous phases, improving the mechanical performance of the cement-based materials.

Further, it was verified that there are correlations between the amorphous phases in the cement-based materials with the mechanical properties investigated in the present work, as well as with the water absorption capacity, where the best performance in these properties was verified for the samples that presented large proportions of amorphous phases. The quantification of the amorphous phases detected from XRD

is necessary to identify the effects of different mineral additions on pozzolanic reactions in the extension to the hydration of cement-based materials overtime in the formation of important phases, such as the CSH phases (2θ between 28° and 33°).

Finally, an important aspect is the filler effect, which is responsible for large dispersions of the cement grains in the fresh state, increases the surface contact of the anhydrous phases with the water [36]. This effect occurs due to the interaction of the amorphous phase of the additions to the phase of the portlandite formed in the hydration of C_3S and C_2S systems. Elevated levels of the mineral additions can consume the portlandite present in the cementitious matrix and increase the possibility of carbonation of the cement-based material, which tends to reduce the useful life of the structure. Therefore, mineral additions act through pozzolanic reactions in the refinement of the microstructure, where secondary CSH crystals decrease the average pore diameter of the cementitious matrix. However, this phenomenon does not occur in an expressive and singular pathway that is associated with the filler effect. By considering the filler effect, the mineral additions, in general, improve the properties of concrete, in particular the pozzolanic additions, including properties associated with the durability of reinforced concrete structures.

3.4. Water absorption capacity and microstructural aspects

By employing the ASTM C1585 standard [37], the behavior of the water absorption of the cement-based materials can be evaluated. Fig. 4 shows the average values of the water absorption capacity of the materials studied in the present article after 3, 6, 24, 48, and 72 h.

From Fig. 4 the samples of the cement-based reference exhibited a more water absorption capacity (an average of 3.6% after 72 h) when compared with the cement-based materials prepared with mineral additions of fly ashes, metakaolin, and silica fume, that exhibited values of water absorption capacity after 72 h of 2.45%, 2.03%, and 1.37%, respectively. These values of the ability of the materials to absorb water can be rationalized as the capacity of their components, which are essentially composed of oxide and hydroxide compounds, to interact with water molecules. Recently, it was reported that the nature of this chemical interaction is the formation of hydrogen bonds [24]. The hypothesis of the different abilities of the materials to absorb water, seemingly, is due to their porosities, which a large quantity of these structural defects tends to enable the formation of hydrogen bonds between the inorganic components of the cement-based materials (mainly

Table 3

Rietveld refinement data of the cement-based materials.

Cement-based material	#Code ^a	Mineral Phase	Compound	Quantity (%)
Reference	44–1481	Portlandite	Ca(OH) ₂	25.30
	41–1451	Ettringite	Ca ₆ Al ₂ (SO ₄) ₃ (OH) ₁₂ ·26H ₂ O	14.51
	05–0586	Calcite	CaCO ₃	10.98
	49–1673	Calcium silicate	Ca ₂ SiO ₄	3.87
	33–0306	Calcium silicate hydrate	Ca _{1.5} SiO _{3.5} •xH ₂ O	9.68
			/1.5CaO·SiO ₂ •xH ₂ O	
	41–0219	Calcium aluminum oxide carbonate hydrate	Ca ₄ Al ₂ O ₆ CO ₃ ·11H ₂ O	2.69
			/3CaO·Al ₂ O ₃ ·CaCO ₃ ·11H ₂ O	
	14–0698	Sorosilicate mineral	Al ₁₃ Si ₅ O ₂₀ (OH, F)18Cl	5.06
	–	Amorphism	–	27.91
Silica Fume	44–1481	Portlandite	Ca(OH) ₂	9.61
	41–1451	Ettringite	Ca ₆ Al ₂ (SO ₄) ₃ (OH) ₁₂ ·26H ₂ O	14.01
	14–0698	Sorosilicate mineral	Al ₁₃ Si ₅ O ₂₀ (OH, F)18Cl	4.29
	05–0586	Calcite	CaCO ₃	10.18
	49–1673	Calcium silicate	Ca ₂ SiO ₄	5.24
	33–0306	Calcium silicate hydrate	Ca _{1.5} SiO _{3.5} •xH ₂ O	13.47
			/1.5CaO·SiO ₂ •xH ₂ O	
	41–0219	Calcium aluminum oxide carbonate hydrate	Ca ₄ Al ₂ O ₆ CO ₃ ·11H ₂ O	2.01
			/3CaO·Al ₂ O ₃ ·CaCO ₃ ·11H ₂ O	
	–	Amorphism	–	41.19
Metakaolin	14–0698	Sorosilicate mineral	Al ₁₃ Si ₅ O ₂₀ (OH, F)18Cl	5.31
	41–0219	Calcium aluminum oxide carbonate hydrate	Ca ₄ Al ₂ O ₆ CO ₃ ·11H ₂ O	3.12
			/3CaO·Al ₂ O ₃ ·CaCO ₃ ·11H ₂ O	
	33–0306	Calcium silicate hydrate	Ca _{1.5} SiO _{3.5} •xH ₂ O	11.30
			/1.5CaO·SiO ₂ •xH ₂ O	
	49–1673	Calcium silicate	Ca ₂ SiO ₄	6.06
	05–0586	Calcite	CaCO ₃	10.81
	41–1451	Ettringite	Ca ₆ Al ₂ (SO ₄) ₃ (OH) ₁₂ ·26H ₂ O	14.55
	44–1481	Portlandite	Ca(OH) ₂	9.68
	86–1560	Quartz	SiO ₂	0.62
Fly Ashes	–	Amorphism	–	37.55
	14–0698	Sorosilicate mineral	Al ₁₃ Si ₅ O ₂₀ (OH, F)18Cl	5.58
	41–0219	Calcium aluminum oxide carbonate hydrate	Ca ₄ Al ₂ O ₆ CO ₃ ·11H ₂ O	2.69
			/3CaO·Al ₂ O ₃ ·CaCO ₃ ·11H ₂ O	
	33–0306	Calcium silicate hydrate	Ca _{1.5} SiO _{3.5} •xH ₂ O	12.50
			/1.5CaO·SiO ₂ •xH ₂ O	
	49–1673	Calcium silicate	Ca ₂ SiO ₄	3.97
	05–0586	Calcite	CaCO ₃	11.34
	41–1451	Ettringite	Ca ₆ Al ₂ (SO ₄) ₃ (OH) ₁₂ ·26H ₂ O	15.96
	44–1481	Portlandite	Ca(OH) ₂	11.66
	86–1560	Quartz	SiO ₂	0.38
	–	Amorphism	–	35.92

^a The reference codes can be accessed at crystallographic letters PDF (Powder Diffraction File).

calcium hydroxide) and water molecules. When the mineral additions are performed, the porosity of the materials, in general, is reduced, which tends to decrease the probability of the hydrogen bond formation between the material and water [24]. The samples prepared with silica fume preserve the characteristics of the sand on their surface, which can lead to the formation of weak hydrogen bonds when compared with other cement-based materials, such as concretes [24]. The values of water absorption capacity measured for the cement-based materials with additions of metakaolin and fly ashes can be rationalized by the quantity of SiO₂ of these mineral additions (~64% for metakaolin, and 70% of the fly ashes). When compared with silica fume, which is 96% of SiO₂ (Table 1), other components present on the surface of both metakaolin and the fly ash cement-based materials, such as calcium hydroxide, are capable of interacting more efficiently with water molecules via hydrogen bonds [24]. Therefore, the higher ability of the main components of the cement-based materials of metakaolin and fly ashes to interact with water molecules when compared with the material of silica fume is, apparently, an important aspect of the nature of the capacity of the cement-based materials to absorb water.

In addition, the specific surface areas of the mineral systems were 1.80 m²/g, 11.16 m²/g, 16.91 m²/g, and 20.42 m²/g, for Portland cement, fly ashes, metakaolin, and silica fume, respectively. The specific areas through the chemical effect and the reactivity of the particles can be correlated with the modified Chapelle method, where both silica fume and metakaolin exhibited the largest I_{PA} values (1498 mgCaO/g

for silica fume and 1195 mgCaO/g for metakaolin).

In order to evaluate aspects of the surface of the materials, Fig. 5 shows the SEM images of all ingredients, and cement-based materials.

From Fig. 5 the morphology of the Portland cement is not homogeneous, and the material presents agglomerated grains that can be formed because of the temperature employed in its processing (1450 °C). The morphology of the silica fume is composed of spherical clusters. Similar aspects were verified for the metakaolin, whose clusters were identified by SEM experiments. However, the morphology of the fly ashes is not well defined when compared with the silica fume. The particles of the fly ashes are massive and well-defined spheres with varying sizes. The SEM images of the cement-based materials, the formation of hydrated cement phases, such as the ettringite in the reference with its well-defined sticks was verified. For the cement-based materials with the addition of silica fume, the formation of the CSH phase is identified with its “honeycomb” and “flower/leaf” shapes. For the samples with metakaolin, the calcium hydroxide presence is well-defined by its morphology in a hexagonal shape. For the cement-based material prepared with fly ashes, all phases were identified. These results showed the more selectivity of phases when the silica fume is employed, whereas the usage of the fly ashes lead to the possibility of more diversity of phases according to their pozzolanic activities. In particular, the CSH phase composes 50%–60% of cement-based materials [38]. By considering an atomic/molecular level, the CSH phase is formed by layers of SiO₂ chains that can be replaced by aluminum compounds, such as the Al₂O₃. Besides, another important

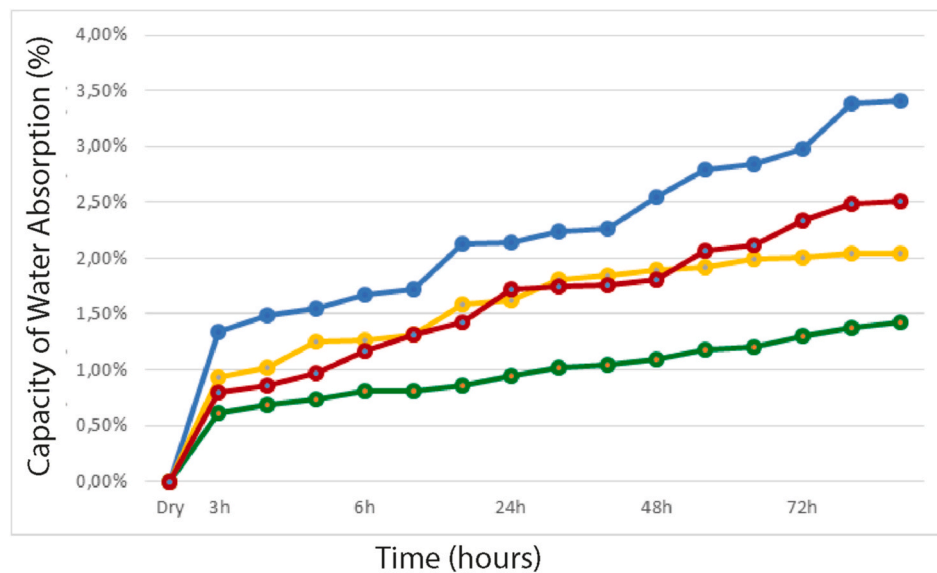


Fig. 4. The average of the values of the water absorption by considering both dried and saturated samples of the four types of cement-based materials (reference (blue), and with additions of silica fume (green), metakaolin (yellow) and fly ashes (red)). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

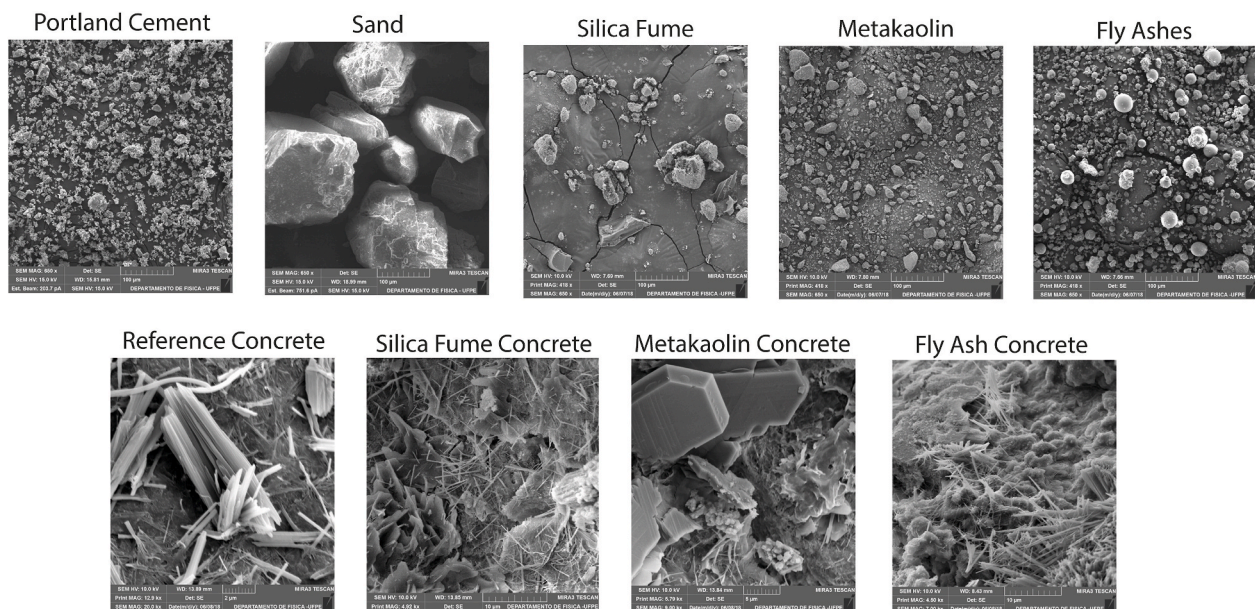


Fig. 5. SEM images of the precursor solid materials and of the cement-based materials. The images were obtained with evaporated gold particles and resolution between 650x and 20 kx magnification.

system that can be present in the final step of the hydration process of cement is the portlandite. This system presents a characteristic shape of hexagonal prismatic crystals, constituting about 20%–25% of the volume of solids in the hydrated cement paste. The formations of these crystals are affected by the availability of space, the temperature at which the cement is being hydrated, among other factors that act directly on this formation.

From the energy-dispersive x-ray spectra associated with the surface of the cement-based materials, the proportion of calcium is similar (an average of 35.5%). However, both silicon and aluminum elements are different for these materials. For example, the sample of the reference has 25.7% of silicon, whereas the cement-based material with silica fume presents 40.2% of silicon. The larger proportion of silicon can be rationalized as the higher quantity of the SiO_2 in the surface of the

material, leading to the low ability to the cement-based material to absorb water. The cement-based materials with additions of metakaolin and fly ashes have 37.6% and 39.2% of the silicon in their samples. These values indicate that the surface of the materials prepared with mineral additions is similar, and, therefore, their corresponding capacities to absorb water are similar. However, the low proportion of silicon dioxide on the surface of the reference can be associated with its more ability to interact with water molecules, which can lead to more capacity for the material to absorb water.

From Fig. 6, all samples are focused on the possible region of the *CSH* phase due to dense morphology with the approximate shape of a “honeycomb”. The SEM/EDS data revealed that for all samples, the dense formation of the *CSH* phase was formed. The Ca/Si values are 1.23 for the reference, 0.88 for the silica fume, 0.95 for the metakaolin, and

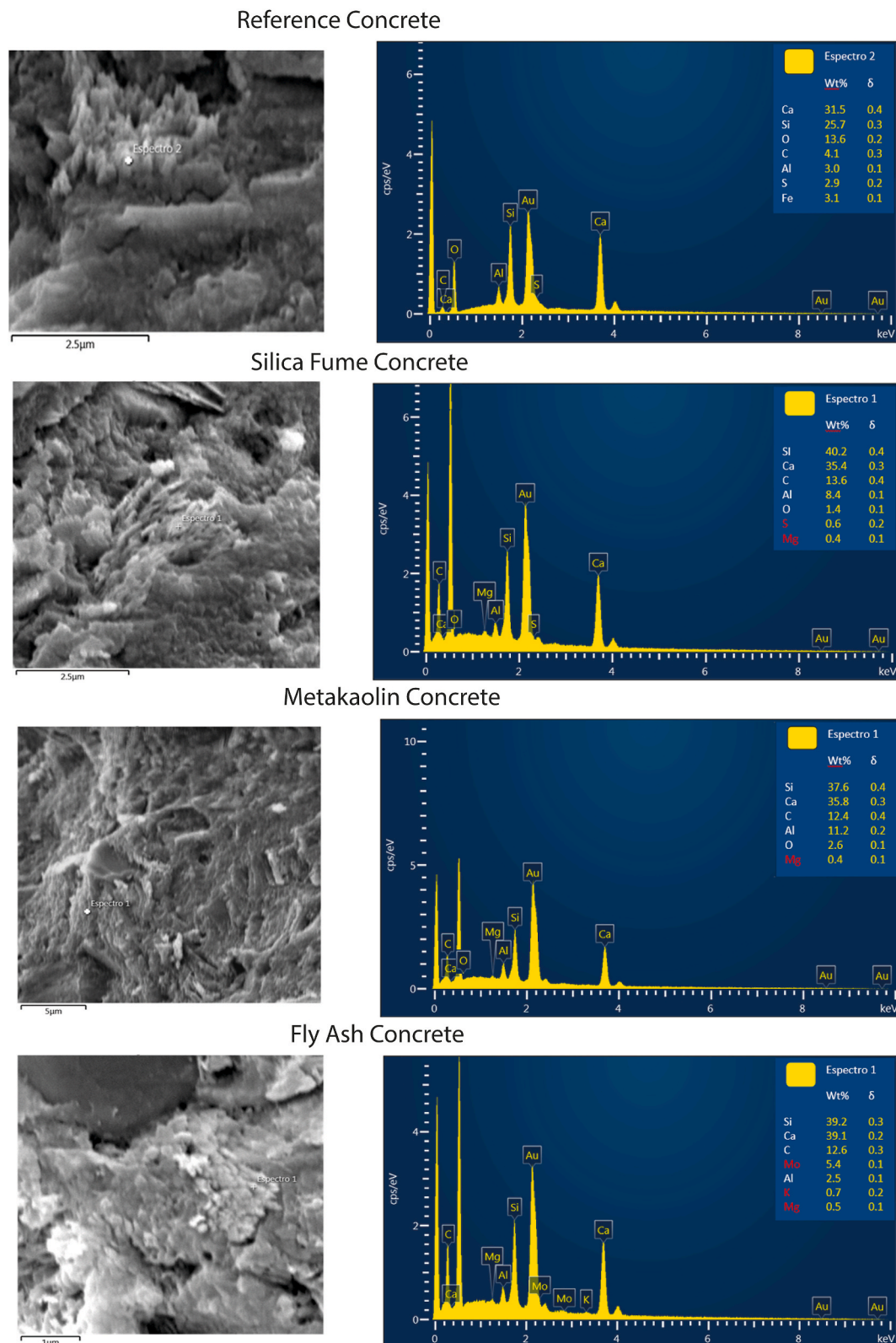


Fig. 6. SEM images and energy dispersive x-ray spectra of the cement-based materials.

0.99 for the fly ash cement-based materials. Therefore, when the addition minerals are considered in the processing of the cement-based materials, the ratio between calcium and silicon in the surface of the materials, are seemingly associated with their ability to absorb water molecules.

On the other hand, the pozzolanic activity modifies the

microstructure of the cement matrix through chemical reactions between the Portlandite and the pozzolan system, which is a compound formed by stable, hydrated products, such as the *CSH*. This phenomenon led to the refinement of the pores of the cement-based materials, which is associated with the refinement of the microstructure due to the formation of *CSH*.

Table 4

Average values of the axial compressive strength of the cement-based materials by considering the following curing times: 28, 63, 91, and 365 days. For each value, six samples were considered.

Cement-based material	Axial compressive strength average (MPa)			
	28 days	63 days	91 days	365 days
Reference	45.45	48.01	50.14	58.53
Silica fume	72.04	71.67	75.09	73.63
Metakaolin	62.71	64.95	59.72	62.86
Fly ash	63.29	65.42	65.01	63.34

Table 5

Average values of the axial tensile resistance of the cement-based materials by considering the following curing times: 28, 63, 91, and 365 days. For each value six samples were considered.

Cement-based material	Tensile resistance average (MPa)			
	28 days	63 days	91 days	365 days
Reference	5.04	5.34	5.13	5.47
Silica fume	7.11	7.17	7.51	6.53
Metakaolin	6.27	6.50	5.97	6.29
Fly ash	6.33	6.54	6.50	6.33

3.5. Mechanical properties and chemometrics analyses

Tables 4 and 5 show the average values of the axial compressive strength and tensile resistance properties of all cement-based materials

studied in the present article.

From Table 4 all cement-based materials prepared with mineral additions exhibited an increase of the axial compressive strength, performance when compared with the reference. For example, by considering a curing time of 28 days, both cement-based materials with metakaolin and fly ash were increased, respectively, by 38% and 39% when compared to the reference. A large increase of 59% was obtained when the silica fume was employed. An increase of this mechanical behavior was also verified over a year, when, after 365 days, the axial compressive strength average of the reference was increased by 26%, 7%, and 8%, when silica fume, metakaolin, and fly ash were employed.

On the other hand, from Table 5, the tensile resistance average of the samples of the reference was also increased. For example, after 63 days, the addition of the silica fume, metakaolin, and fly ash increased the tensile resistance of the reference by 34%, 22%, and 22%, when silica fume, metakaolin, and fly ash were employed. After 365 days, the corresponding effects on the tensile resistance of the reference were 19%, 15%, and 16%, when silica fume, metakaolin, and fly ash were employed. By considering both axial compressive strength and tensile resistance of the cement-based materials, the silica fume led to the largest mechanical performances when compared to the behaviors exhibited for the cement-based materials with metakaolin and fly ash additions. Besides, the formation of the CSH and the physical effect from the refinement of pores are observed through capillary absorption and mechanical resistance.

In order to elucidate more aspects of the impact of the presence of the mineral additions considered in this article, a chemometric analysis was carried out. The mechanical properties' experiments were performed in

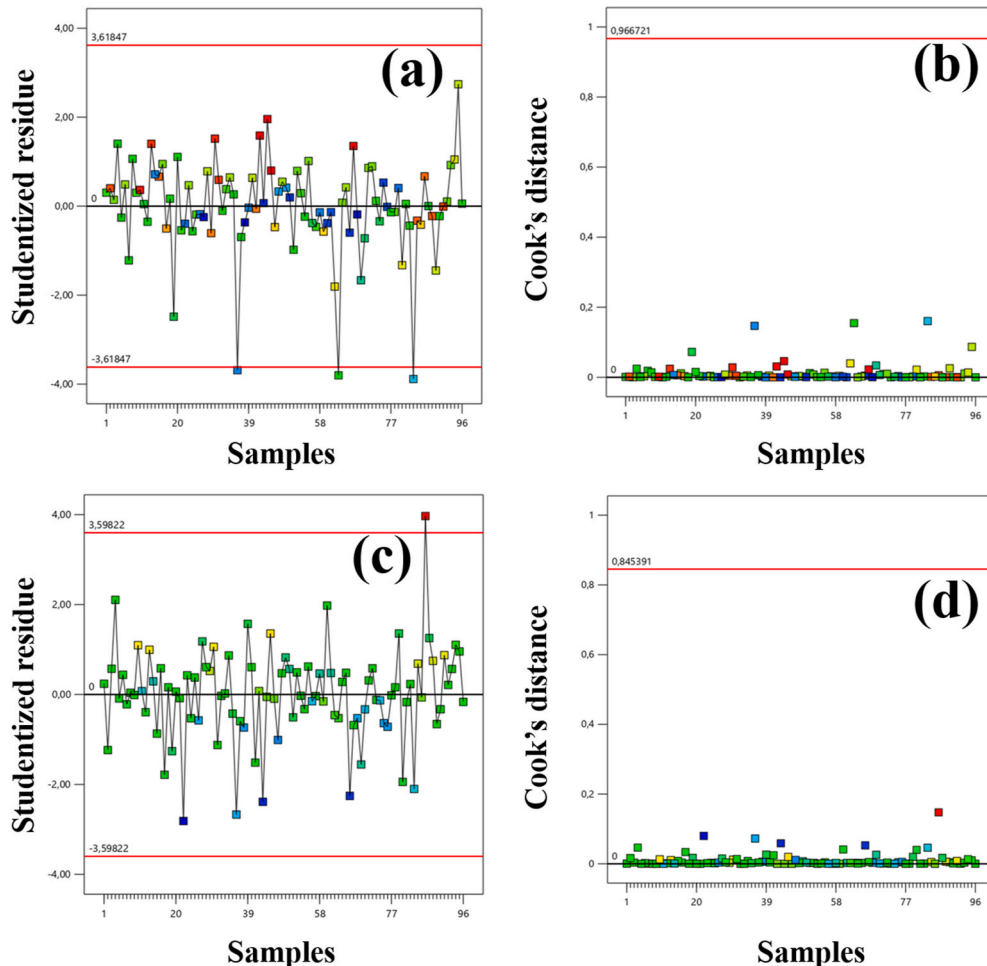


Fig. 7. Studentized residue and Cook's distance were respectively calculated for axial compressive strength (a, b) and tensile resistance (c, d).

a systematic manner when each one of the four cement-based materials was evaluated in four different times by considering a total of 96 samples (Tables S4 and S5 of the Supplementary Material). These samples of the cement-based materials were submitted to axial compressive strength and tensile resistance tests. By using the measured values, two models were built, one for each type of test, that was individually submitted to the analysis of variance (ANOVA) statistical test. The hypothesis tests were calculated for each model using F critical values and also the mean square relation between the parameter and the model error. The chemometric analysis indicated that the parameters of time and the interaction between time and the type of cement-based material strongly affect the axial compressive strength resistance values.

From the ANOVA evaluation, only the significant parameters were included in the model, whose Studentized residues can be visually inspected in Fig. 7. Three samples were over the limits established by the t-test for the residue of the model built for axial compressive strength, and another for the tensile resistance. These samples are indicated as potential outliers, but when Cook's distance was analyzed, no outlier is present in the model. Therefore, no sample was excluded from the built model. This situation is common in t-tests when the overall standard deviation of the model is small, which increase the standardized residue values like the studentized residue. A homoscedastic profile is observed, which indicates that both models are well adjusted.

From Fig. 8 (a) and 8(b), the silica fume cement-based material exhibited both higher axial compressive and tensile resistance highlighted by the built model. Therefore, when both responses were

evaluated in the desirability function (Fig. 8 (d)), the silica fume cement-based material also presented the largest mechanical properties. Further, Fig. 8 (c) shows that the cement-based reference (●) increases the resistance within the experimental time, while the other cement-based materials are more invariable. The silica fume cement-based material (● in Fig. 8 (c)) exhibited the largest mechanical properties in 91 days, resulting in 75.09 ± 2.54 MPa and 7.08 ± 0.20 MPa values for the axial compressive strength and tensile resistance, respectively. By considering the four analyzed cement-based materials, while silica fume presents the largest mechanical properties and the reference the lowest, both materials with metakaolin and fly ash present intermediary resistances and similar behaviors within the experimental time.

4. Conclusions

The XRF measurements together with the physico-chemical reasons indicate that the strong pozzolanic activity index of the silica fume measured by the modified Chapelle's method is likely due to its large ability to form the calcium silicate hydrated system when compared with both metakaolin and fly ash additions that exhibit lower values of pozzolanic activities. The main difference of the pozzolanic activity of the silica fume and the metakaolin and fly ashes is seemingly due to the possibility of formation of more diverse phases when both metakaolin and fly ash are considered; whereas, for the silica fume, only the calcium silicate hydrated system is preferentially formed. Therefore, strict control of the purity of the material always needs to be performed to obtain

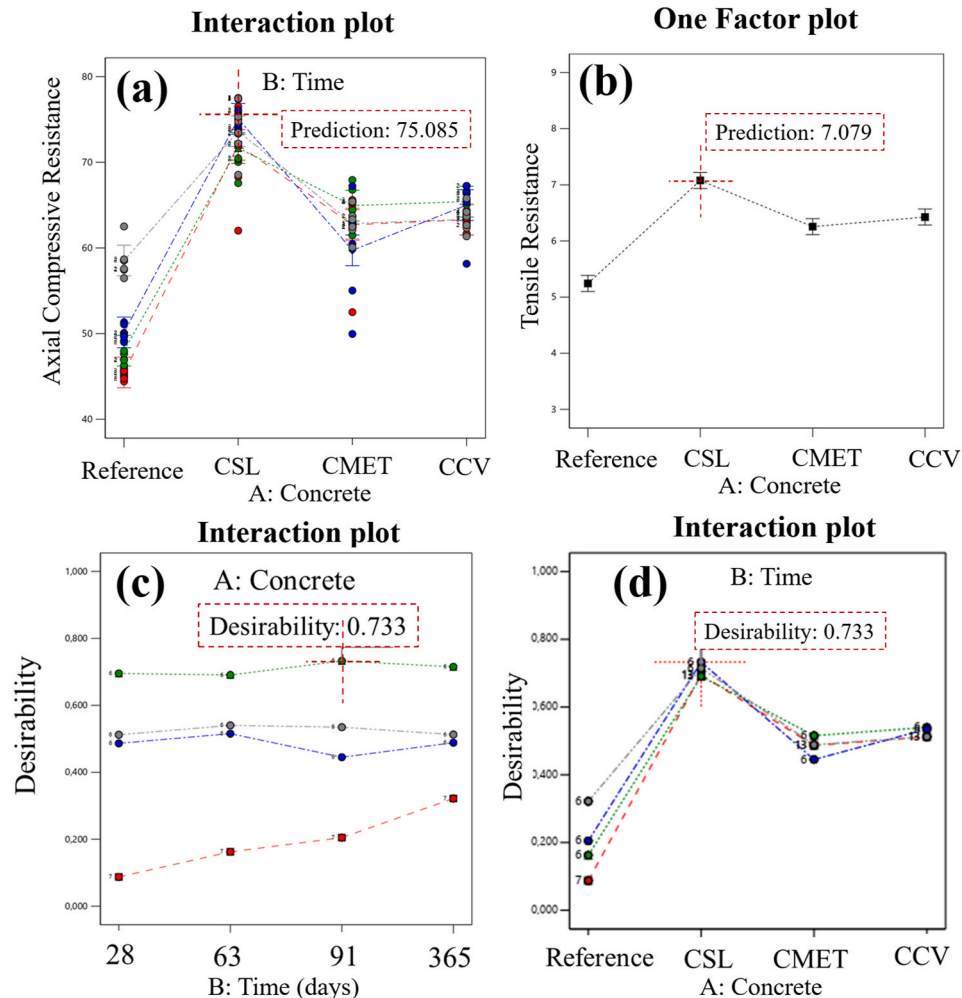


Fig. 8. Interaction plot for axial compressive resistance (a), the one factor plot for tensile resistance (b), the desirability interaction plot for time (c) and cement-based material (d).

mineral systems with powerful pozzolanic activities.

On the other hand, the values of the thermodynamic properties of the hydration reactions of the cement-based materials revealed that the mineral additions considered are not capable of affecting, in a significant manner, the hydration of the Portland cement when the cement-based materials were prepared.

The XRD results revealed that the presence of the mineral systems increases the formation of the amorphous CSH phase. This aspect was mainly verified for the silica fume, which presented more amorphous characteristics, and reacted in an efficient manner with the portlandite phase. The ability of the silica fume to react with portlandite phase leads to large pozzolanic activities when compared with both metakaolin and, especially, with the fly ashes.

Further, SEM/EDS results showed that the surface of the cement-based materials prepared with mineral additions is similar and have a large proportion of SiO₂ when compared with the cement-based reference. The low proportion of SiO₂ on the surface of the cement-based reference can be associated with its ability to interact more efficiently with the water molecules, leading to a material with a larger water absorbing capacity.

Finally, chemometric analyses showed that the cement-based material with silica fume exhibited the largest mechanical properties in 91 days, resulting in 75.09 ± 2.54 MPa and 7.08 ± 0.20 MPa values for the axial compressive strength and tensile resistance, respectively. By comparing all cement-based materials, the materials prepared with silica fume exhibited the largest mechanical properties and the cement-based reference the lowest, both cement-based materials with metakaolin and fly ashes present intermediary resistances and similar behaviors.

CRedit authorship contribution statement

R. Berenguer: Formal analysis, Writing – original draft. **N. Lima:** Formal analysis, and. **L. Pinto:** Formal analysis, performed the Chemometric Analysis. **E. Monteiro:** Formal analysis, performed the experiments of the Modified Chapelle's method and analyzed the data. **Y. Povoas:** performed the Mechanical Properties experiments and XRF Spectrometry. **R. Oliveira:** Formal analysis, Writing – original draft, and. **N.B.D. Lima:** Formal analysis, Writing – original draft, performed the experiments of the XRD, Thermogravimetric Analysis, Experiments of water Absorption Capacity and SEM/EDS, designed the study, analyzed all the results, and wrote the article.

Declaration of competing interest

The authors declare that there are no conflicts of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.job.2021.102358>.

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